

Event System Pro - Application Rules

Plain-language summary of rules, restrictions, and constraints for the Event System Pro front-end and related database/demo-data conventions.

General layout and navigation

- Main content blocks (messages, buttons, fields, dropdowns) are capped at 360px wide and centered.
- Default route is Login (/). Unknown URLs redirect to login.
- Public routes: /, /register, /forgot-password.
- Protected routes require a signed-in session (see Authentication and Roles).
- Staff reaches Judging via Staff -> Contest, which goes to /judging.
- Judging Submit and Back to Staff both return to /staff.

Authentication and session

- Login calls PostgREST RPC api.login with username or email and password.
- Passwords verified with bcrypt on the server.
- On success: session in sessionStorage (esp_session) with user_id, username, email, roles, JWT token.
- Authenticated API calls send Authorization: Bearer <token>.
- Logout clears session and returns to /.
- Sign in again after DB auth or role changes so JWT includes current roles.

Roles and route guards

Seven roles: admin, staff, judge, headjudge, registration, floorcoordinator, competitor.

- ProtectedRoute: no session -> /; missing role -> /home.
- Admin role passes any role check.
- /home, /account, /changepassword: any signed-in user.
- /staff, /judging: staff or admin.
- /competitor: competitor or admin.
- /adminhome, /admin/*: admin only.

Home page (/home)

- After login, user lands on Home (role hub).
- Buttons: Staff (/staff), Competitor (/competitor), Admin (/adminhome), Account (/account).
- Skin dropdown (ThemeSwitcher) below navigation buttons.
- Back to Login logs out and returns to /.
- Test Messages is on Admin home, not Home.

Admin section

- Admin home: Event Details, Contests, Competitors, Competition Entries, Staff, Generate Attendees.
- Generate Attendees calls api.generate_demo_attendees (admin JWT); regenerates demo attendee_id 1-3000.
- Test Messages on Admin home shows one success, warning, and problem alert.
- Event Details and Competition Entries are placeholders.
- Contests page navigates to /admin/contests/contest (results).
- Admin competitors: paginated user table; 25 rows desktop, 10 mobile.

Competitor page (/competitor) - placeholder

- Opened from the Competitor button on Home.
- Page title: Competitor (ContestSelectionPage with title="Competitor").
- Three full-width buttons: Contest 1, Contest 2, Contest 3.
- Contest buttons are placeholders - they do not navigate anywhere (no contestRoute).
- Back to Home button navigates to /home.
- Same layout shell as Staff but without a working contest destination yet.

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Message boxes

App-wide stacked alerts at the top of the screen (not Judging-specific). Wrapped by MessageProvider in main.tsx.

Types and appearance

- Three message types: success, warning, and problem.
- Each type uses a distinct outlined MUI Alert color scheme (green, amber, red).
- Fixed at top center (top: 16px), capped at 360px wide (CONTENT_MAX_WIDTH).
- Container uses aria-live: polite for screen readers.

Stacking and dismissal

- Multiple messages can be visible at the same time, stacked vertically.
- Messages do not auto-dismiss - they persist until the user clicks one.
- Clicking a message plays a collapse animation (~350ms), then removes it.
- New messages slide in from above when added.

API and usage

- Any page can call useMessages() (must be inside MessageProvider).
- Show messages with showSuccess(text), showWarning(text), or showProblem(text).
- clearMessages() removes all messages immediately (no exit animation).
- dismissMessage(id) removes a single message immediately (after collapse animation).

Page-specific behavior

- Login, Register, and Forgot password call clearMessages() on mount.
- Admin home has a Test Messages button that clears the stack, then shows one of each type:
 - Success: Your change has been saved.
 - Warning: Your event starts in less than 15 min.
 - Problem: Your sign in time has passed.

Theme / skin switcher

App-wide MUI theming via skins. AppThemeProvider wraps the app in main.tsx.

Home page placement

- Home page (/home) exposes the skin control via ThemeSwitcher.
- Layout order: Staff and Competitor buttons, then Skin dropdown, then Test Messages and Back to Login.
- Dropdown uses the same 360px centered width as other main controls.

Dropdown behavior

- MUI Select with label "Skin" (FormControl, size small, full width).
- Two options: Light and Dark.
- Changing the selection updates the active skin immediately across the entire app.

Persistence

- Selected skin id saved to localStorage key event-system-pro.skin-id.
- On load, stored value is applied; invalid or missing values fall back to Light.
- Legacy stored value default is treated as Light.

App-wide application

- AppThemeProvider builds MUI theme from active skin and supplies ThemeProvider + CssBaseline.
- All pages inherit the current skin.
- useThemeSwitcher() exposes skinId, setSkin, skins, and currentSkin.

Judging page - competitor data

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- Each session shows 20 couples (accordion rows).
- Each couple has a bib number (1-999, unique within the set, sorted ascending), a Leader, and a Follower (first + last name).
- Names come from Legion of Super-Heroes (DC) and Shi'ar Imperial Guard (Marvel-inspired) pools.
- Each character has a sex value (male or female) based on common comic portrayals.

Name selection rules

- Leader: pick a male character 90% of the time.
- Follower: pick a female character 90% of the time.
- The other 10% can be anyone still available in the pool.
- A character may appear only once in the whole dataset, in either Leader or Follower slot.
- Names are excluded if first or last name contains any whole word: boy, girl, lad, lass, man, woman (also ladd).

Accordion behavior

- Only one panel can be open at a time.
- Collapsed row (accordion title): bib # | leader swatch * names * follower swatch | score.
- Leader color cell (when set) is to the left of the leader name.
- Follower color cell (when set) is to the right of the follower name.
- Names between swatches: Leader * Follower.
- Expanded row shows: Raw Score first, then leader and follower rows (with color picking).
- Clicking inside expanded details does not collapse the accordion.

Sort/filter freeze while editing

- While a panel is open, the list does not re-sort or re-filter, even if scores change or the sort/filter dropdown changes.
- Re-sort/re-filter happens when the user collapses the open panel or opens a different panel.
- Changing the sort/filter dropdown always closes any open panel and applies the new option immediately.

Raw scores

- Label is "Raw Score".
- Entered with 4 digit dropdowns (0-9), displayed as XX.XX (e.g. 87.43).
- Score range: 0.00 to 99.99.
- Summary score area is blank until the score has been touched/changed.
- Once any digit changes, the score shows in the collapsed header immediately.
- A couple counts as scored only if raw score is greater than 0.0 (not merely touched at 0.00).

Assign Random Scores

- Dropdown action (not a persistent sort mode).
- Sets every couple to a random score between 30.0 and 99.9.
- Marks all as touched, switches sort to Sort by Raw Score, and closes any open panel.

Sorting and filtering

Dropdown options:

- Sort by Bib #: ascending bib number (default).
- Sort by Raw Score: highest score first; unscored/touched-at-zero at bottom; bib # breaks ties.
- Sort by Leader's Last Name: A-Z; bib # breaks ties.
- Sort by Follower's Last Name: A-Z; bib # breaks ties.
- Unscored Only: hides couples with a non-zero raw score.
- Assign Random Scores: bulk random scoring (see above).

Additional sort rules

- When 100% complete is first reached, sort automatically switches to Sort by Raw Score.

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- Unscored Only treats a couple as unscored if never touched, or touched but still ≤ 0.0 .

Progress and submit

- Top of page shows Percent Complete bar (replaces old "Judging" header).
- Progress = couples with non-zero raw score / 20 x 100.
- Text: Percent Complete: X%.
- At 100%, the bar becomes a Submit button.

Duplicate raw scores

When an accordion closes (collapse, switch panels, or change sort/filter while one was open), if that couple's raw score matches another couple's score (> 0), open the Duplicate Score dialog.

Dialog rules

- Title: Duplicate Score.
- Close X in top-right (dismisses without changing scores).
- Two choices using leader and follower first names: put the current couple higher, or put the other couple higher.

Resolution logic

- Preferred fix: raise the chosen couple's score by +0.1.
- If +0.1 would collide with someone else's score: leave the chosen couple alone and lower the other duplicate couple by 0.1 instead.
- Scores stay within 0.00-99.99.
- After a choice, if sort is Sort by Raw Score, the list re-sorts to reflect new scores (including when another panel is still open).

Color picking

- Each leader and follower can have two colors: top and bottom.
- Colors are chosen from a fixed palette (8 columns x 8 rows).
- Palette source: front-ends/front-end-cursor/src/data/judgingColorPalette.ts.
- Color picker dialog width matches accordion content (360px max).
- Palette icon opens the dialog if no colors yet; swatch opens it if colors exist.
- First click sets top color; second click sets bottom color, then auto-save and close.
- Titles: Pick Top Color for {firstname}, then Pick Bottom Color {firstname}.
- X cancels without saving.
- Reopening shows prior selections; first click replaces top, second replaces bottom.
- After one pick in a session, clicking the backdrop saves that one color and closes.
- Color swatches appear in the collapsed summary when at least one color is set.
- Colors are stored per bib + role (leader or follower).
- Hover a swatch in the dialog to see its HEX and RGB designation.

Color palette (8 x 8)

Rows are listed top to bottom; columns left to right. Row 1 uses fixed hex values. Rows 2-7 are hue gradients (stored as HSL in code); HEX and RGB below are rendered equivalents. Row 8 is a fixed neon assortment (HSL in code).

Row 1 - Neutral (black to white)

Col 1: #000000 rgb(0, 0, 0)
Col 2: #1f1f1f rgb(31, 31, 31)
Col 3: #3d3d3d rgb(61, 61, 61)
Col 4: #5c5c5c rgb(92, 92, 92)
Col 5: #7a7a7a rgb(122, 122, 122)
Col 6: #999999 rgb(153, 153, 153)
Col 7: #d9d9d9 rgb(217, 217, 217)
Col 8: #ffffff rgb(255, 255, 255)

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Row 2 - Blue gradient

Col 1: #193a56 rgb(25, 58, 86)
Col 2: #1d4a76 rgb(29, 74, 118)
Col 3: #205898 rgb(32, 88, 152)
Col 4: #2264bb rgb(34, 100, 187)
Col 5: #236ede rgb(35, 110, 222)
Col 6: #3e7ce7 rgb(62, 124, 231)
Col 7: #5a8cf0 rgb(90, 140, 240)
Col 8: #789ef6 rgb(120, 158, 246)

Row 3 - Purple gradient

Col 1: #3c1f5a rgb(60, 31, 90)
Col 2: #502477 rgb(80, 36, 119)
Col 3: #662895 rgb(102, 40, 149)
Col 4: #7f29b5 rgb(127, 41, 181)
Col 5: #9a2ad5 rgb(154, 42, 213)
Col 6: #b041e0 rgb(176, 65, 224)
Col 7: #c359e9 rgb(195, 89, 233)
Col 8: #d573f1 rgb(213, 115, 241)

Row 4 - Red gradient

Col 1: #4d181e rgb(77, 24, 30)
Col 2: #6b1d24 rgb(107, 29, 36)
Col 3: #8c1f27 rgb(140, 31, 39)
Col 4: #ae2027 rgb(174, 32, 39)
Col 5: #d21f23 rgb(210, 31, 35)
Col 6: #e62e2e rgb(230, 46, 46)
Col 7: #ef4d48 rgb(239, 77, 72)
Col 8: #f66b63 rgb(246, 107, 99)

Row 5 - Brown gradient

Col 1: #3a2a21 rgb(58, 42, 33)
Col 2: #4d3627 rgb(77, 54, 39)
Col 3: #60432c rgb(96, 67, 44)
Col 4: #744f31 rgb(116, 79, 49)
Col 5: #8a5d34 rgb(138, 93, 52)
Col 6: #a16c36 rgb(161, 108, 54)
Col 7: #b87b37 rgb(184, 123, 55)
Col 8: #cb8c3d rgb(203, 140, 61)

Row 6 - Green gradient

Col 1: #1e471f rgb(30, 71, 31)
Col 2: #246229 rgb(36, 98, 41)
Col 3: #297f34 rgb(41, 127, 52)
Col 4: #2c9e40 rgb(44, 158, 64)
Col 5: #2dbe4d rgb(45, 190, 77)
Col 6: #36d762 rgb(54, 215, 98)
Col 7: #4ce27d rgb(76, 226, 125)
Col 8: #64eb97 rgb(100, 235, 151)

Row 7 - Yellow gradient

Col 1: #997528 rgb(153, 117, 40)
Col 2: #b78f29 rgb(183, 143, 41)
Col 3: #d6ab2b rgb(214, 171, 43)
Col 4: #e0be41 rgb(224, 190, 65)
Col 5: #e9cf58 rgb(233, 207, 88)
Col 6: #f1dd70 rgb(241, 221, 112)
Col 7: #f7ea8b rgb(247, 234, 139)
Col 8: #fbf4a6 rgb(251, 244, 166)

Row 8 - Neon assortment

Col 1: #fa2d9a rgb(250, 45, 154)
Col 2: #fce620 rgb(252, 230, 32)
Col 3: #1bf513 rgb(27, 245, 19)
Col 4: #07f7c7 rgb(7, 247, 199)
Col 5: #19cdfa rgb(25, 205, 250)
Col 6: #b729f3 rgb(183, 41, 243)
Col 7: #91fc16 rgb(145, 252, 22)

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Col 8: #fa6a2d rgb(250, 106, 45)

Color swatch shape

- Each competitor's colors are shown in a square swatch cell (20x20px).
- Top color fills the top half of the square; bottom color fills the bottom half.
- The cell is split horizontally (top over bottom), not side-by-side.
- In the expanded accordion, the swatch (or palette icon) is on the left of each leader/follower name; name is to the right.
- In the collapsed accordion title, the same split square cells appear when colors are set.
- Leader swatch: immediately to the left of the leader (first) name.
- Follower swatch: immediately to the right of the follower (second) name.
- If only top is set (no bottom yet), top fills the entire square until bottom is chosen.

Name display in summary row

- Names are shown as Leader * Follower in the collapsed accordion title.
- Leader color swatch (when set) is to the left of the leader name.
- Follower color swatch (when set) is to the right of the follower name.
- If space is tight, names shorten progressively: both full -> leader initial -> follower initial -> both initials.
- Shortening accounts for space taken by color swatches.

Mobile-friendly layout

The app is designed to be usable on phones as well as desktop - narrow, centered content rather than wide multi-column layouts.

Viewport and page shell

- index.html: width=device-width, initial-scale=1.0, interactive-widget=resizes-content.
- body uses min-height 100vh with no default margin.

Narrow centered column

- Primary content capped at 360px (CONTENT_MAX_WIDTH) and centered.
- Applies to messages, buttons, fields, dropdowns, and centeredContentStackSx stacks.
- MUI theme caps buttons and form controls at 360px.
- Pages use Container (sm on Login/Register, md elsewhere) with py: 6.
- Buttons in stacks use fullWidth within the centered column.

Touch and small screens

- useIsMobileDevice: coarse pointer, or screen <= 768px and touch-capable.
- Layout uses the narrow column; no separate desktop/mobile page layouts.

Dialogs and overlays

- Modals use Dialog fullWidth maxWidth xs.
- Message stack fixed at top with horizontal padding (px: 2).

Judging on narrow widths

- Accordion summary names shorten when space is tight (ResizeObserver).
- Same accordion UI at all breakpoints; no mobile-only Judging layout.

Mobile and form behavior (Login / Register)

- AppTextField on mobile scrolls focused field into view when keyboard opens.
- Tapping the field container focuses the input.
- Default enterKeyHint: done (single-line) or enter (multiline).
- Default inputMode: text (password fields excluded).

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- Register numeric fields use inputMode: numeric with max lengths on phone/zip fields.
- Outlined inputs use 16px font size to reduce unwanted zoom on focus (especially iOS).

Database audit columns

Every public table uses created_date, created_by, modified_date, modified_by.

- created_date: TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP.
- created_by: varchar(128) NOT NULL.
- modified_date / modified_by: NULL on insert; set on update.
- Do not use epoch placeholder defaults (1970-01-01, etc.).
- Seeds: created_by = c-agent; leave modified_* NULL.
- App users: created_by = username on register; modified_by = username on update.

Demo data and more_json

- event_group, event, and attendee each have more_json (jsonb).
- Demo flag key is "demo" (boolean), not is_demo.
- event_group: seeded with demo true/false.
- event: demo copied from parent event_group.
- attendee: demo copied from parent event on seed insert.
- charter table renamed to club; event.host_charter_id renamed to host_club_id.

Attendee demo data

- attendee_id 1-3000 reserved for demo seed data; production rows must be > 3000.
- Seed selects 5 random demo event_group trios (15 events) x 200 attendees = 3000 rows.
- created_date: 1-90 days before event start_date (same calendar year as event).
- Re-run deletes only attendee_id <= 3000; preserves rows above 3000.
- Admin Generate Attendees calls api.generate_demo_attendees (admin JWT required).
- CLI seed: database/seeds/012_attendee_seed.sql calls generate_demo_attendees_core.
- PostgREST needs PGRST_JWT_SECRET matching database app.jwt_secret for authenticated RPCs.

Repository layout

- front-ends/front-end-cursor: primary React + Vite + MUI app.
- back-ends/postgres: PostgREST + Swagger + Mailpit + mailer (Docker).
- database/migrations: incremental schema/API; database/seeds: local dev data only.
- scripts/rebuild-local-database.ps1: full local DB rebuild.
- deploy/: Dokploy / production notes.

App shell and provider order

- main.tsx order: AppThemeProvider -> MessageProvider -> AuthProvider -> BrowserRouter -> App.
- ActivityMonitor and LoginFlashHandler render above Routes in App.
- Inactivity timeout: 10 minutes; activity on clicks, form keydown, route changes.
- Server sync: touch_last_activity (30s debounce), session_status (every 15s).
- On expiry: warning message, logout, redirect to /.

Page layout shell

- Container maxWidth md (sm on Login/Register), Paper elevation 3, centered Stack.
- centeredContentStackSx: 360px max width, mx auto.
- Primary buttons: contained fullWidth; back buttons: outlined fullWidth.

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Environment variables (front-end)

- VITE_POSTGREST_URL (default http://localhost:3000).
- VITE_MAILER_URL (default http://localhost:3001).
- VITE_REALTIME_WS_URL (WebSocket POC).

PostgREST API architecture

- HTTP API from PostgreSQL schema api only.
- Roles: anon, authenticated (JWT), authenticator (connection).
- Dev: migration 004 allows anon writes for Swagger.
- Prod: migration 021 revokes anon writes; JWT required for mutations.
- RPCs via POST /rpc/<name>; security definer functions enforce role checks.
- JWT claims: role=authenticated, sub=user_id, username, app_roles array.

Local database development

- rebuild-local-database.ps1: drop DB, baseline, migrations (skip superseded 005-007, 015, 027, 030), seeds 002-012.
- Migrations run in prod; seeds are local dev only.
- configure-local-postgres-trust.ps1: passwordless psql on localhost (dev only).

Fictional demo events

- event_group seeds 008-010 with more_json.demo = true.
- Seed 011 generated by scripts/generate-fictional-event-seed.py.
- Three instances per group: ~2025, ~2026, ~2027 (June anchors).
- 3-5 event days; end_date = start_date + days; event.more_json.demo from group.
- Re-run deletes attendees then events for fictional groups before re-insert.
- api.nightly_cleanup shifts demo event dates forward 1 day (scheduled maintenance).

JSON vs JSONB

- Use jsonb for queryable/mergeable columns (more_json, competitors_json, judges_json).
- Use json for legacy blobs (some user columns, location_json).
- Prefer jsonb for new columns that will be queried.

Staff and contest selection

- StaffPage: ContestSelectionPage title=Staff, contestRoute=/judging.
- CompetitorPage: same layout, no contestRoute (buttons inert).
- Admin contests: three buttons all navigate to /admin/contests/contest.

Lookup tables and owner account

- _lu tables: created_by = c-agent on seed rows; modified_ * NULL.
- dancingevilgenius dev account (seed 004); default password ChangeMeOnFirstLogin!
- Seed 007 grants all seven app roles to dancingevilgenius.

Database migrations

- Files: database/migrations/NNN_description.sql; sorted by prefix.
- Prefer idempotent DDL; NOTIFY pgrst reload schema after API changes.
- Superseded migrations skipped by rebuild-local-database.ps1 when folded into baseline.
- Agent UPDATE in migrations: modified_by = c-agent, modified_date = CURRENT_TIMESTAMP.
- App RPCs use api.set_audit_actor(username) before UPDATE.

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Admin contest results (mock)

- /admin/contests/contest uses buildMockContestResults() (mock data per page load).
- Wide table layout (Container maxWidth lg); Place, Bib, Couple, judge columns.
- Back to Contests returns to /admin/contests.

Technical constraints (implicit but enforced)

- No @mui/icons-material - icons are local SVGs (CloseIcon, PaletteOutlinedIcon, etc.).
- Judging entries are generated once per page load (random bib numbers and names are fixed for that visit).
- Score digit dropdown clicks must not toggle the accordion (stopPropagation).